



BITS Pilani

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CE F231

Fluid Mechanics

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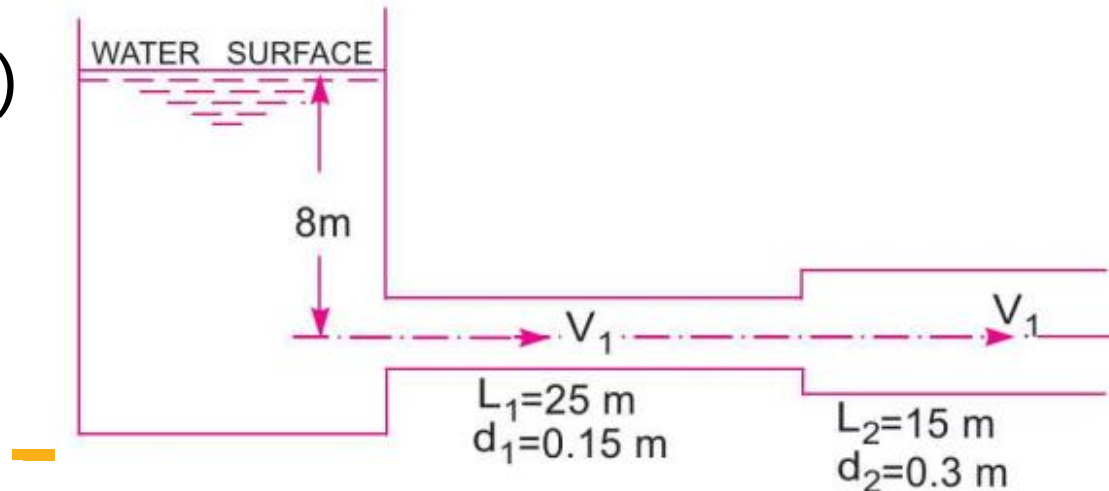
December 5, 2022



Lecture 33: Flow Through Pipes

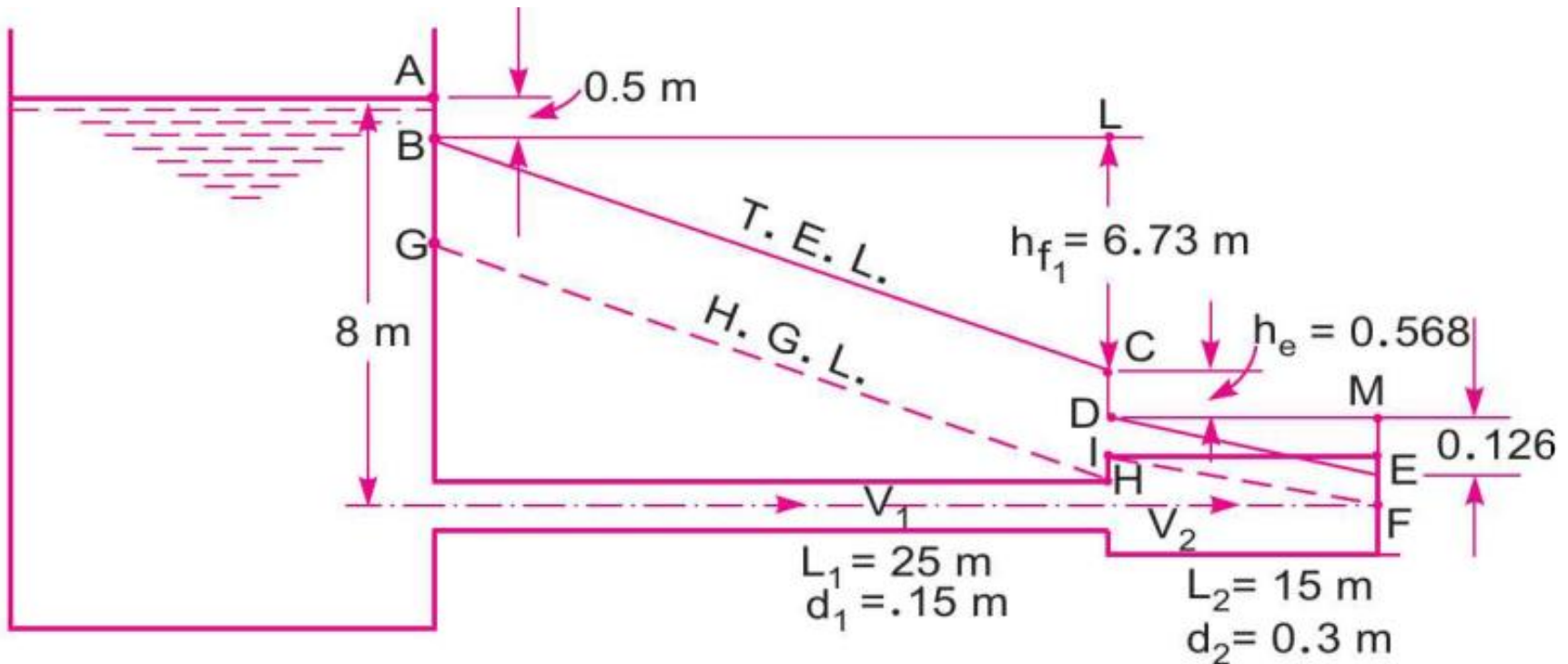
Learning Outcome

- Different types of energy losses in pipes
 - Major losses (occurs due to friction)
 - Hagen-Poiseuille equation
 - Darcy-Weisbach equation
 - Minor losses (due to change in characteristics of the section)



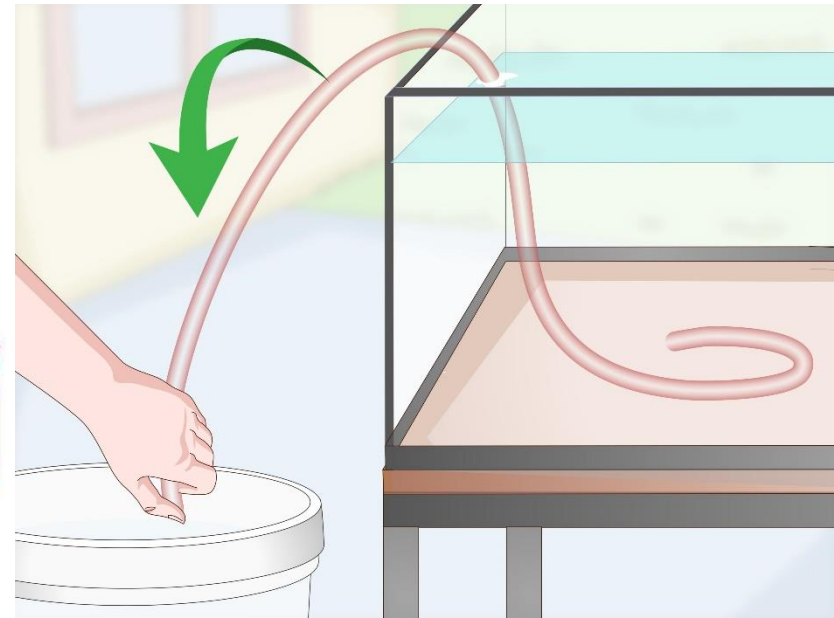
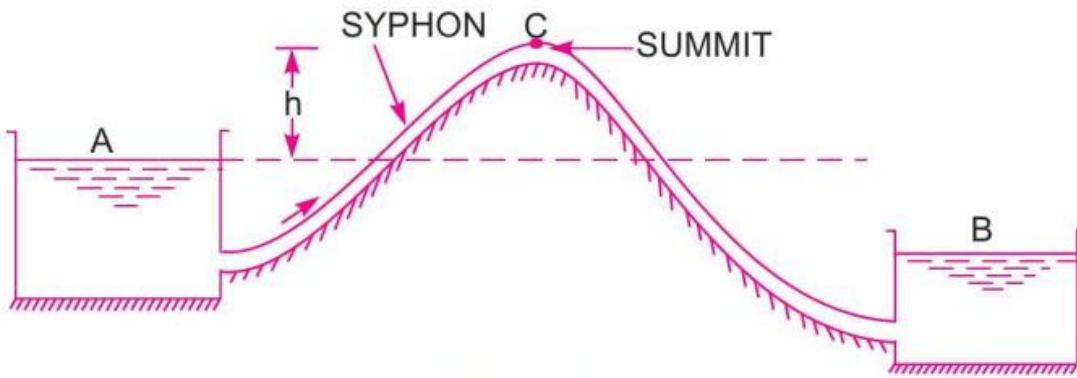
Learning Outcome

➤ Hydraulic gradient line and total energy line



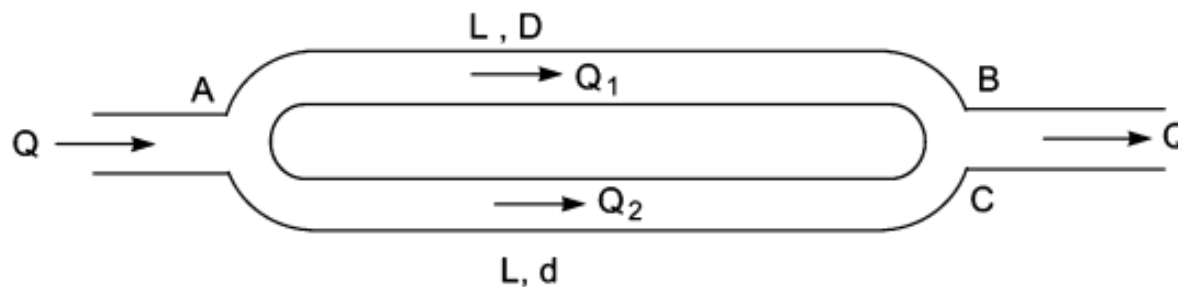
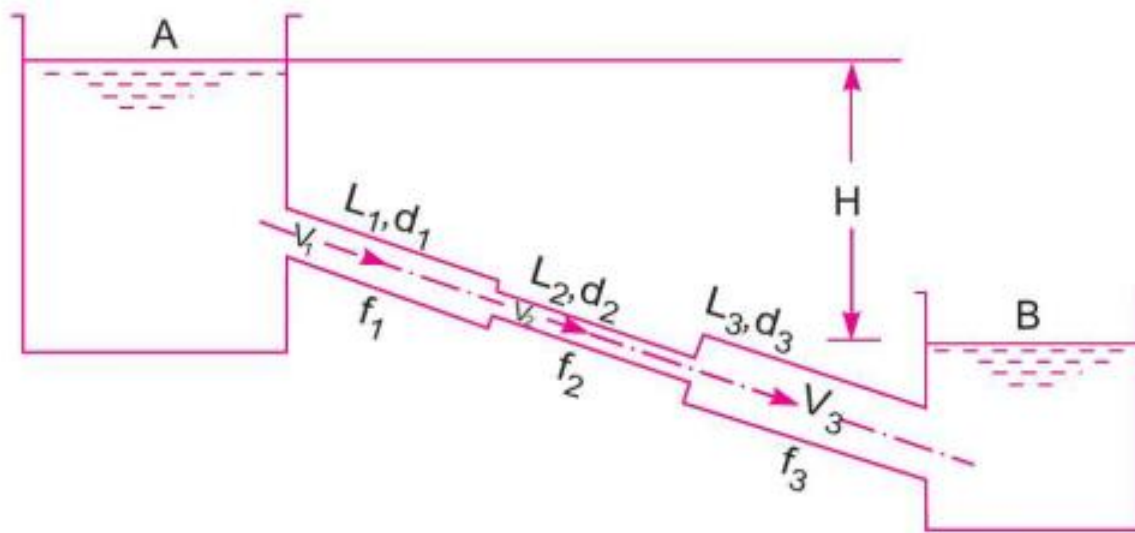
Learning Outcome

➤ Flow through syphon



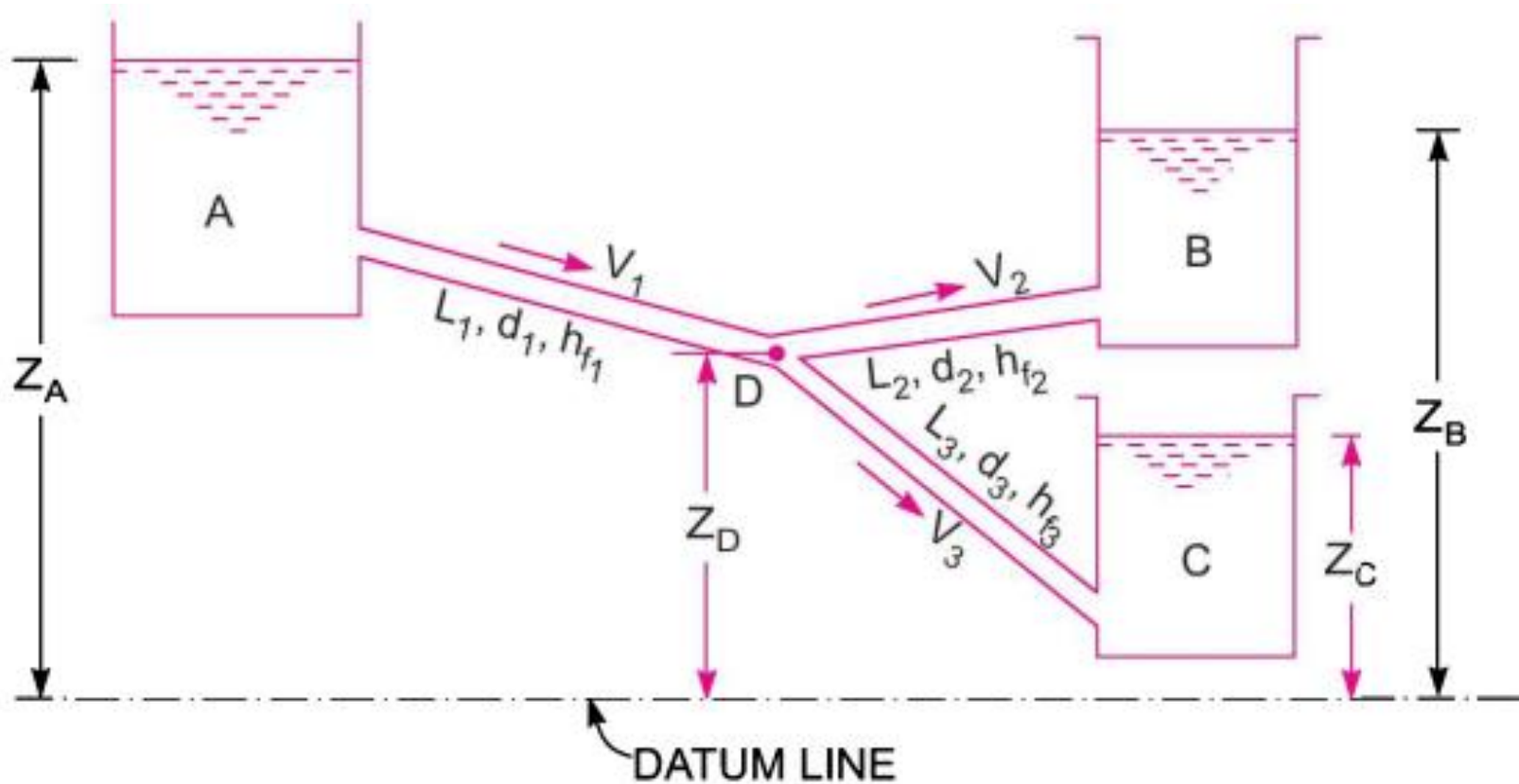
Learning Outcome

- Flow through pipes in series and parallel



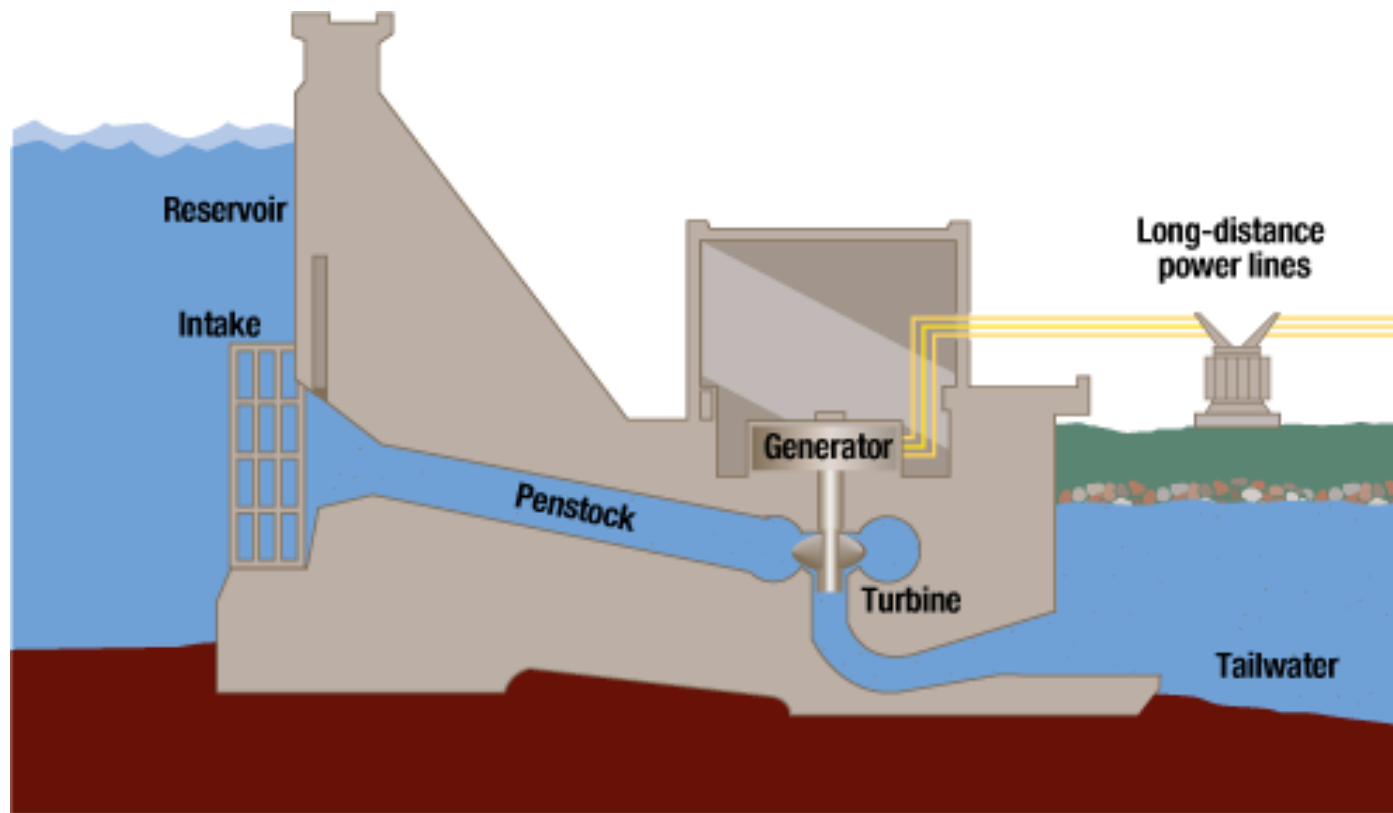
Learning Outcome

➤ Flow through branched pipes



Learning Outcome

➤ Power transmission through pipes



Major losses

➤ Losses due to friction

- Hagen-Poiseuille equation → viscous flow

$$h_f = \frac{32\mu\bar{u}L}{\rho g D^2}$$

- Darcy-Weisbach equation → viscous and turbulent flow

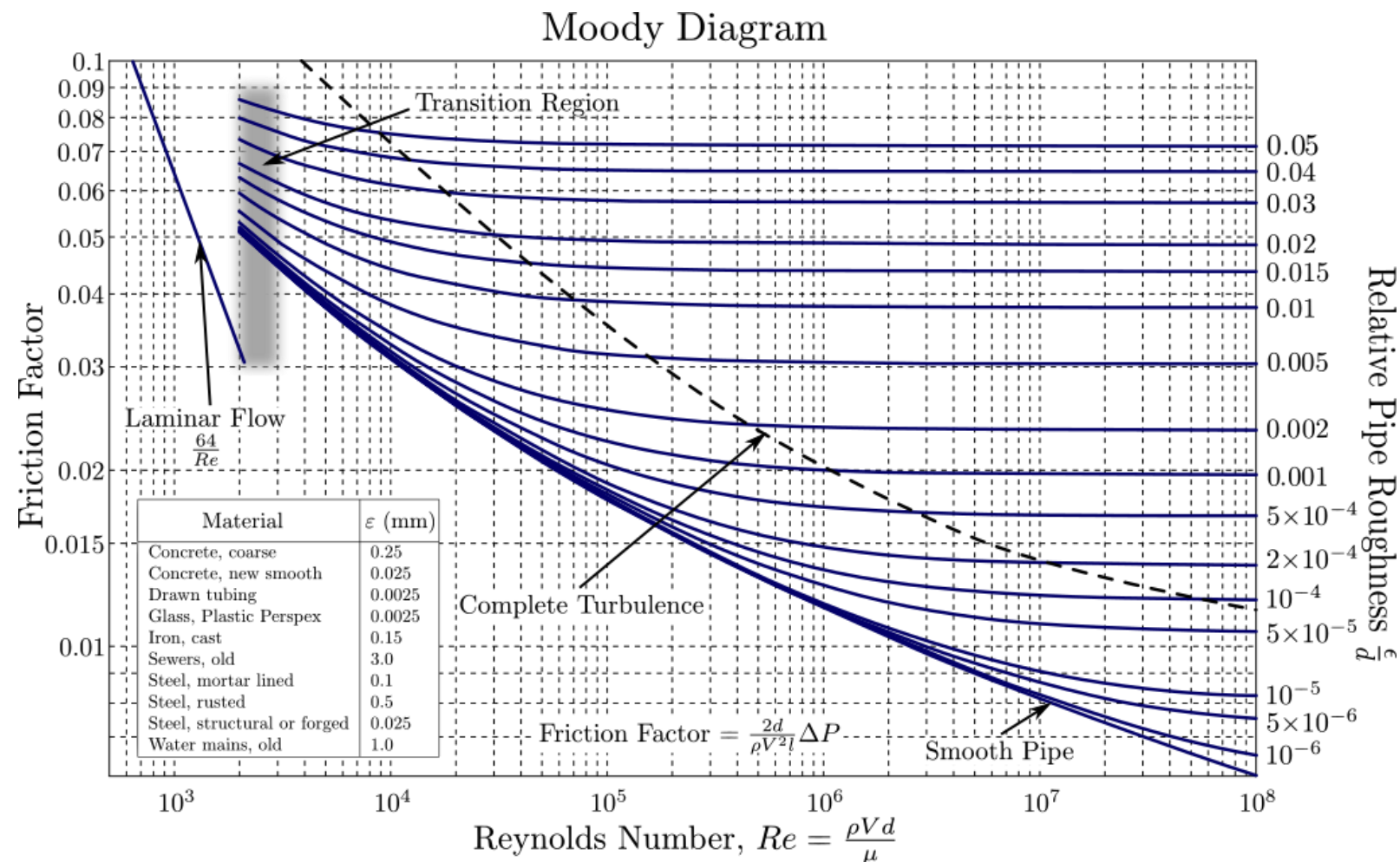
$$h_f = \frac{4 \cdot f'}{2g} \cdot \frac{LV^2}{d} \quad \text{or} \quad h_f = \frac{f \cdot L \cdot V^2}{d \times 2g}$$

f' = coefficient of friction or
Fanning friction factor

f = friction factor

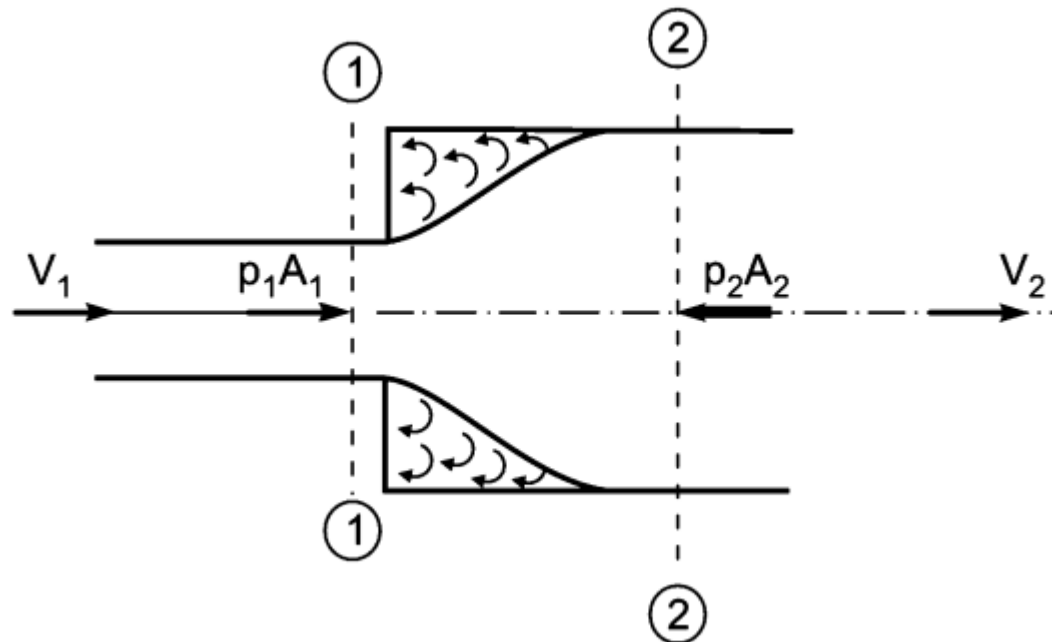
Major losses

➤ Losses due to friction



Minor losses

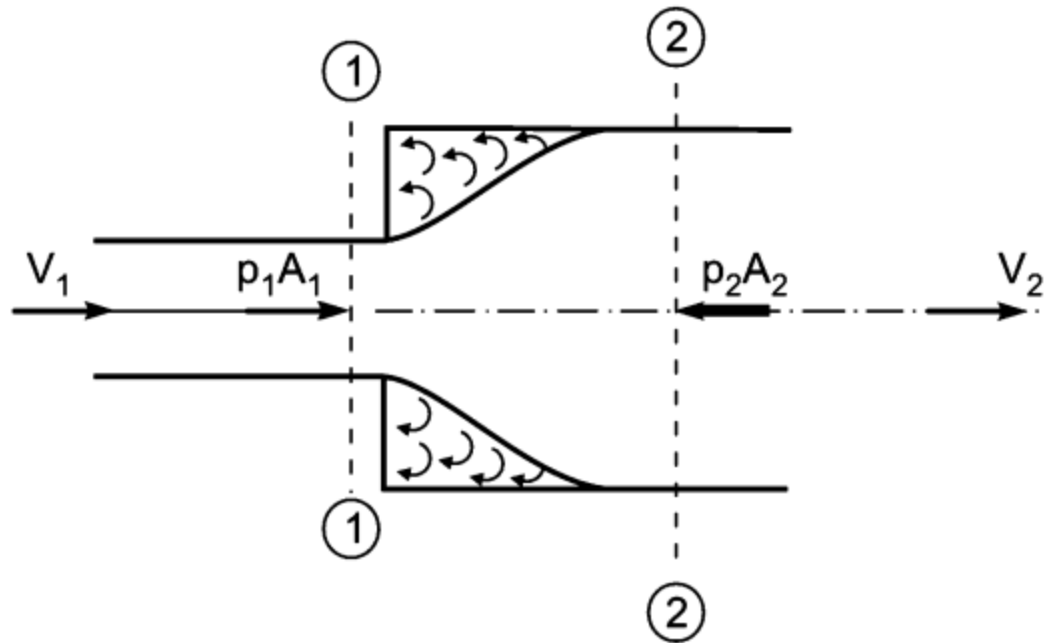
➤ Losses due to sudden enlargement



$$h_e = \frac{(V_1 - V_2)^2}{2g}$$

Minor losses

➤ Losses due to sudden enlargement (derivation)

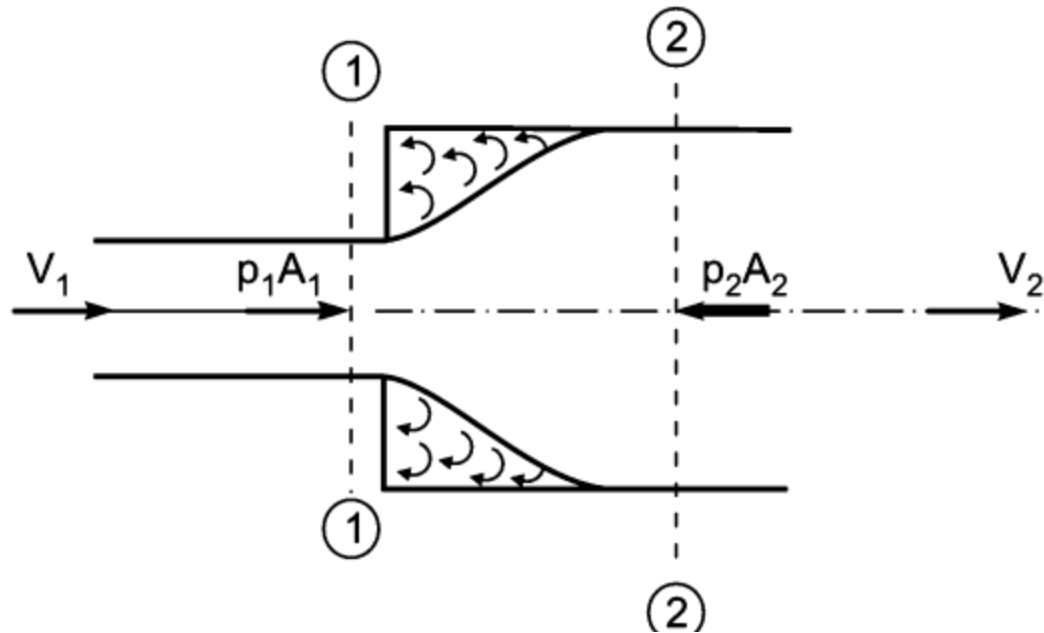


Applying Bernoulli's equation at sections 1-1 and 2-2,

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2 + \text{loss of head due to sudden enlargement}$$

Minor losses

➤ Losses due to sudden enlargement (derivation)

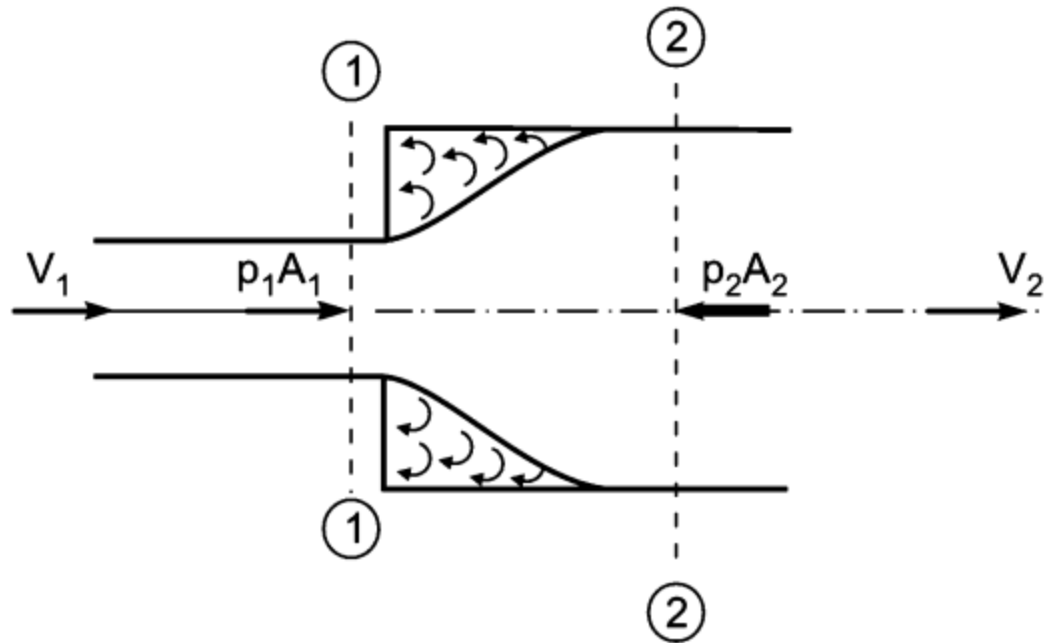


$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + h_e$$

$$h_e = \left(\frac{p_1}{\rho g} - \frac{p_2}{\rho g} \right) + \left(\frac{V_1^2}{2g} - \frac{V_2^2}{2g} \right)$$

Minor losses

➤ Losses due to sudden enlargement (derivation)

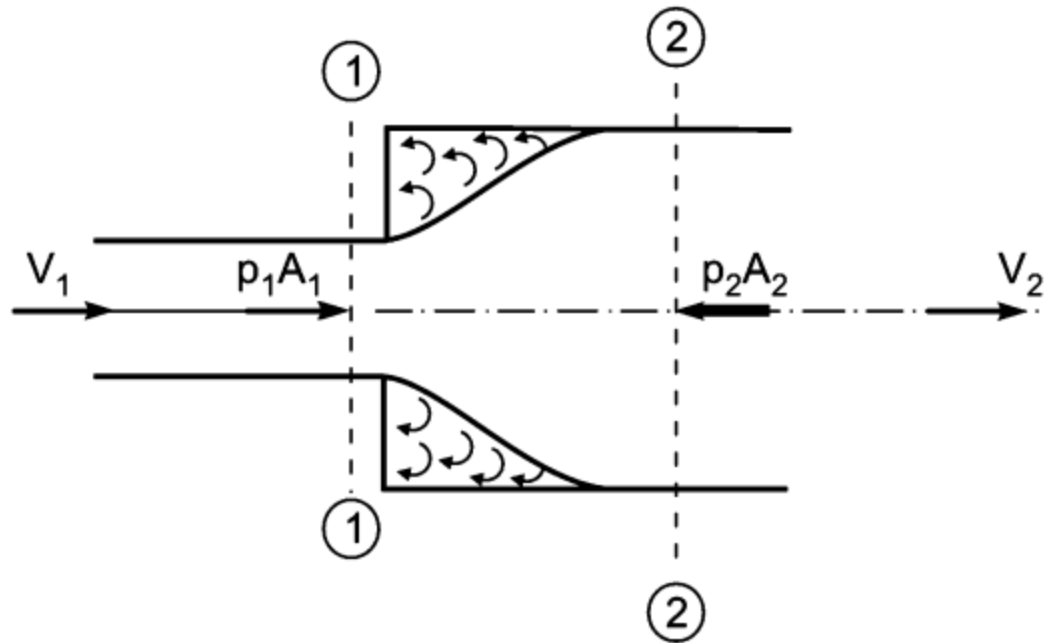


$$h_e = \left(\frac{p_1}{\rho g} - \frac{p_2}{\rho g} \right) + \left(\frac{V_1^2}{2g} - \frac{V_2^2}{2g} \right)$$

Get $p_1 - p_2$ in terms of V_1 and V_2

Minor losses

- Losses due to sudden enlargement (derivation)

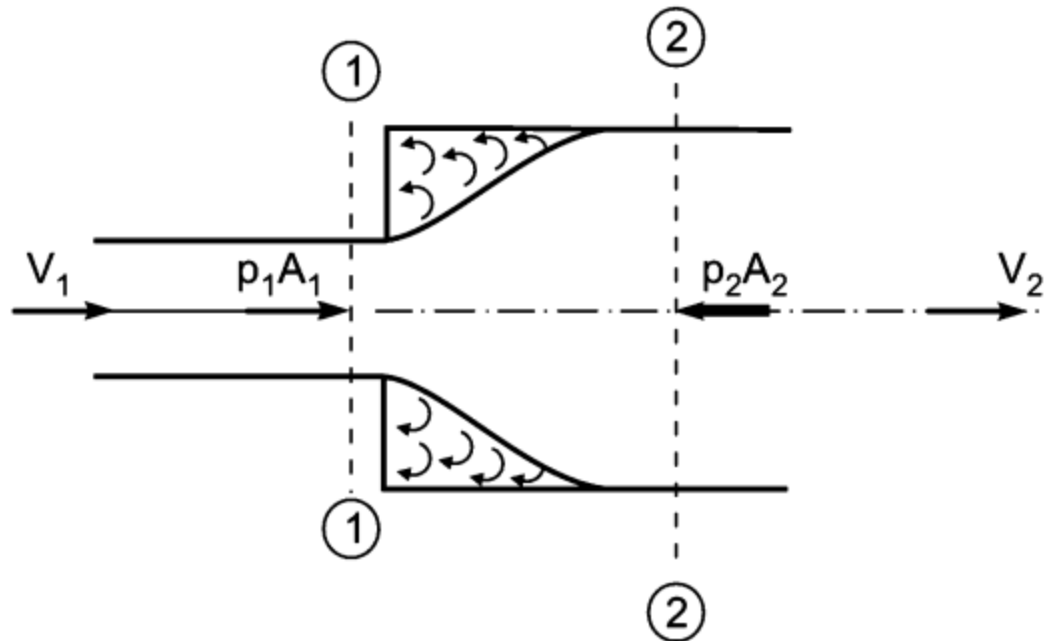


$$F_x = p_1 A_1 + p'(A_2 - A_1) - p_2 A_2$$

$$p' = p_1$$

Minor losses

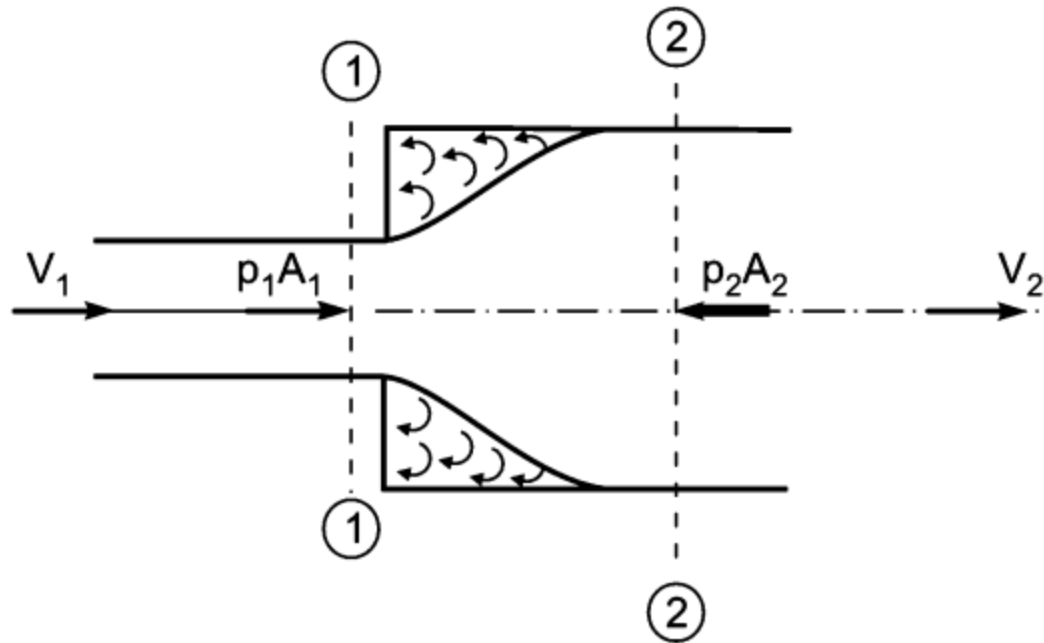
- Losses due to sudden enlargement (derivation)



$$\begin{aligned}
 F_x &= p_1A_1 + p_1(A_2 - A_1) - p_2A_2 = p_1A_2 - p_2A_2 \\
 &= (p_1 - p_2)A_2
 \end{aligned}$$

Minor losses

➤ Losses due to sudden enlargement (derivation)

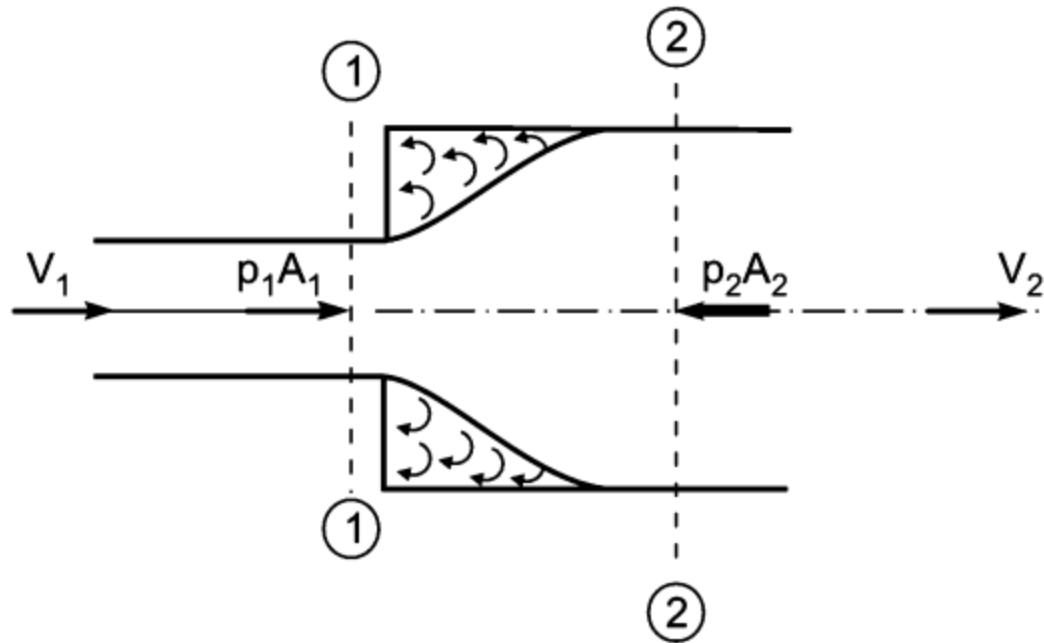


Momentum of liquid/sec at section 1-1 = mass $\times$ velocity

$$= \rho A_1 V_1 \times V_1 = \rho A_1 V_1^2$$

Minor losses

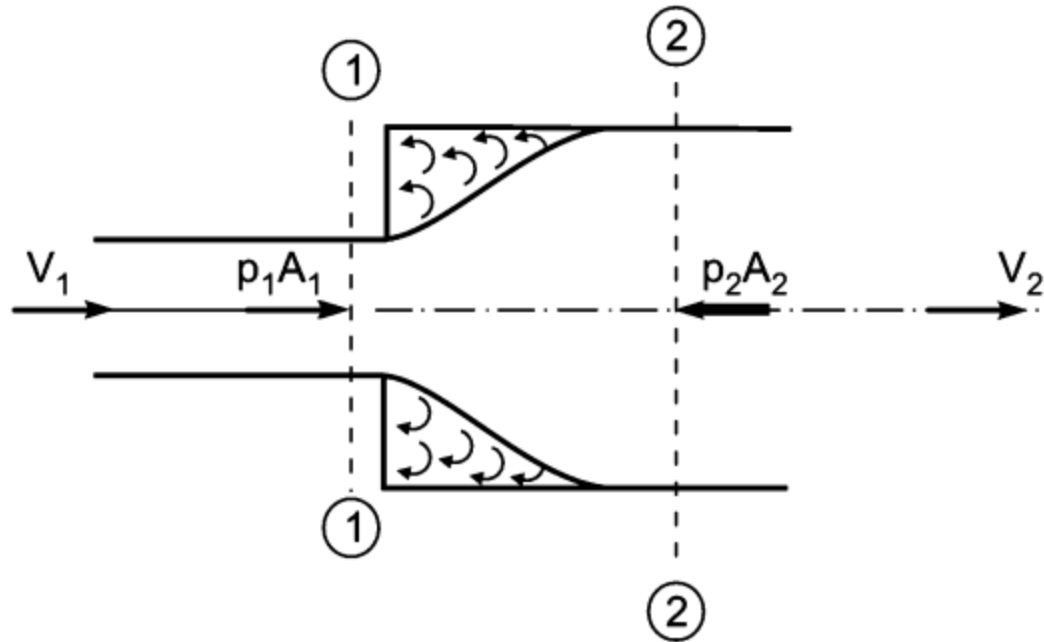
➤ Losses due to sudden enlargement (derivation)



$$\text{Momentum of liquid/sec at section 2-2} = \rho A_2 V_2 \times V_2 = \rho A_2 V_2^2$$

Minor losses

- Losses due to sudden enlargement (derivation)

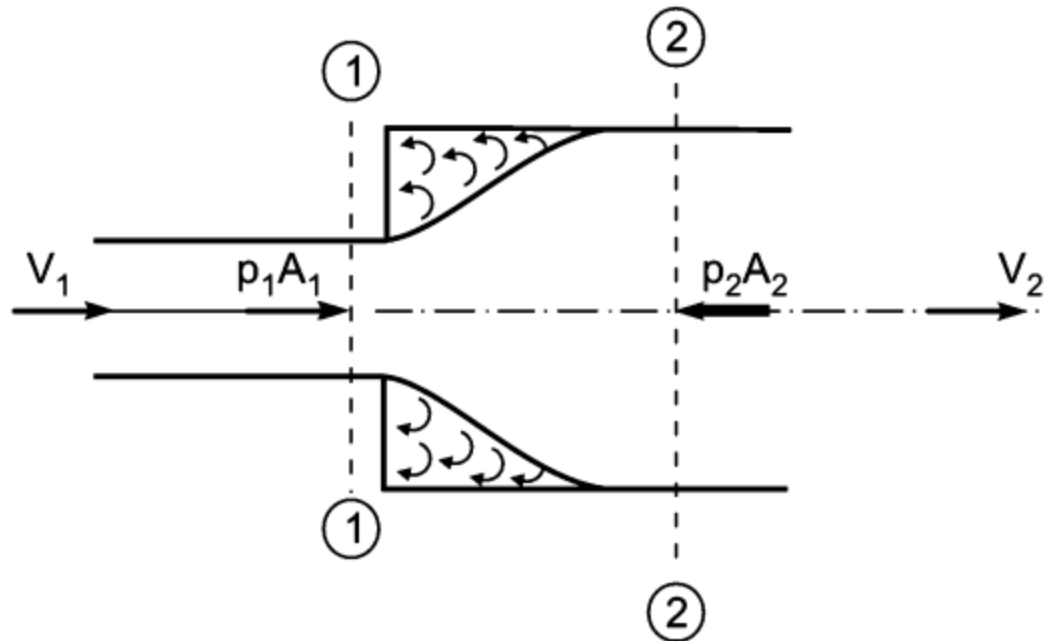


$$\text{Change of momentum/sec} = \rho A_2 V_2^2 - \rho A_1 V_1^2$$

$$A_1 V_1 = A_2 V_2 \text{ or } A_1 = \frac{A_2 V_2}{V_1}$$

Minor losses

➤ Losses due to sudden enlargement (derivation)

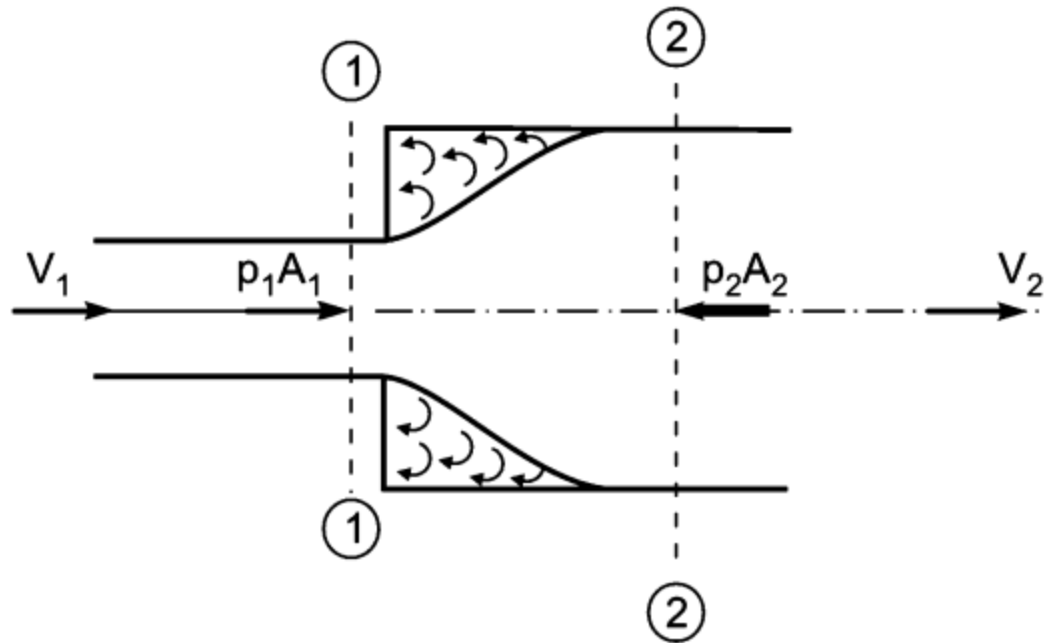


$$\text{Change of momentum/sec} = \rho A_2 V_2^2 - \rho A_1 V_1^2$$

$$\begin{aligned} \text{Change of momentum/sec} &= \rho A_2 V_2^2 - \rho \times \frac{A_2 V_2}{V_1} \times V_1^2 = \rho A_2 V_2^2 - \rho A_2 V_1 V_2 \\ &= \rho A_2 [V_2^2 - V_1 V_2] \end{aligned}$$

Minor losses

➤ Losses due to sudden enlargement (derivation)



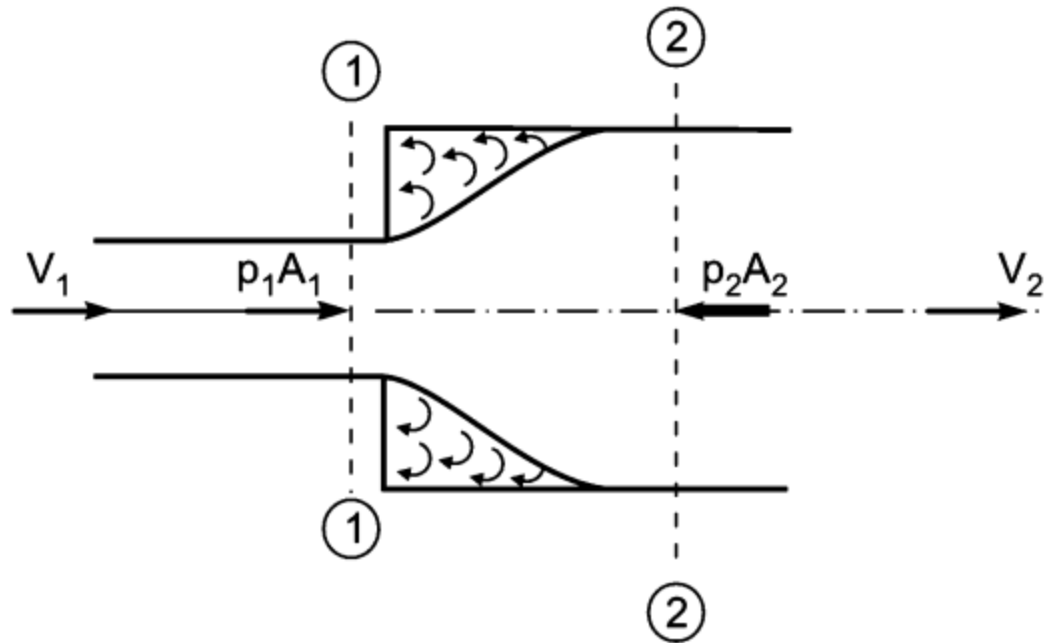
Get $p_1 - p_2$ in terms of V_1 and V_2

$$(p_1 - p_2)A_2 = \rho A_2[V_2^2 - V_1V_2]$$

$$\frac{p_1 - p_2}{\rho} = V_2^2 - V_1V_2$$

Minor losses

➤ Losses due to sudden enlargement (derivation)



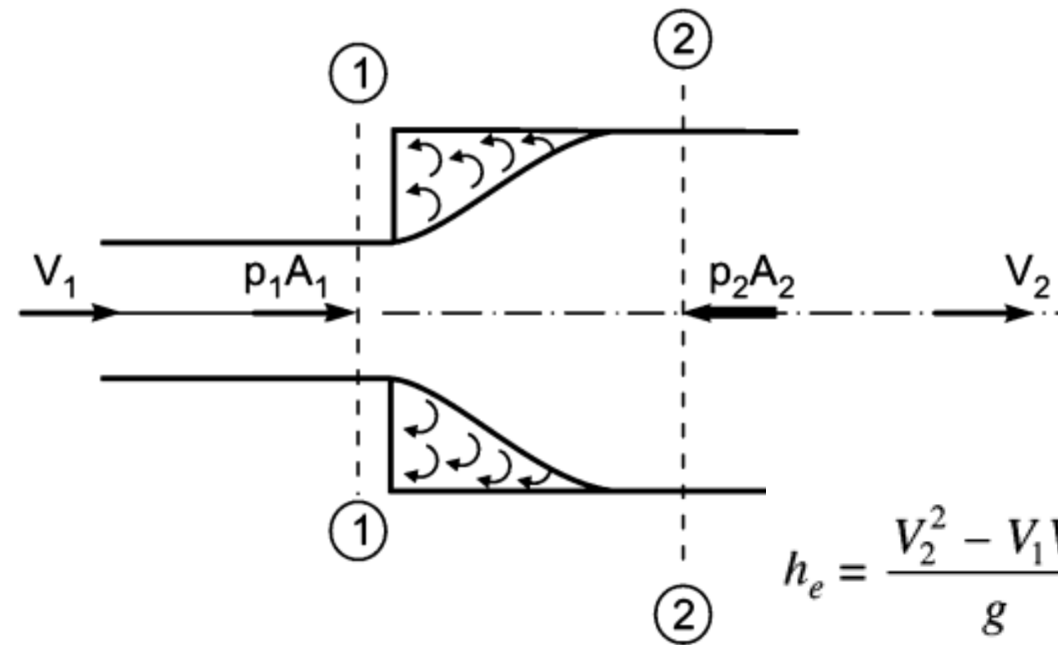
$$(p_1 - p_2)A_2 = \rho A_2 [V_2^2 - V_1 V_2]$$

$$\frac{p_1 - p_2}{\rho} = V_2^2 - V_1 V_2$$

$$h_e = \left(\frac{p_1}{\rho g} - \frac{p_2}{\rho g} \right) + \left(\frac{V_1^2}{2g} - \frac{V_2^2}{2g} \right)$$

Minor losses

➤ Losses due to sudden enlargement (derivation)



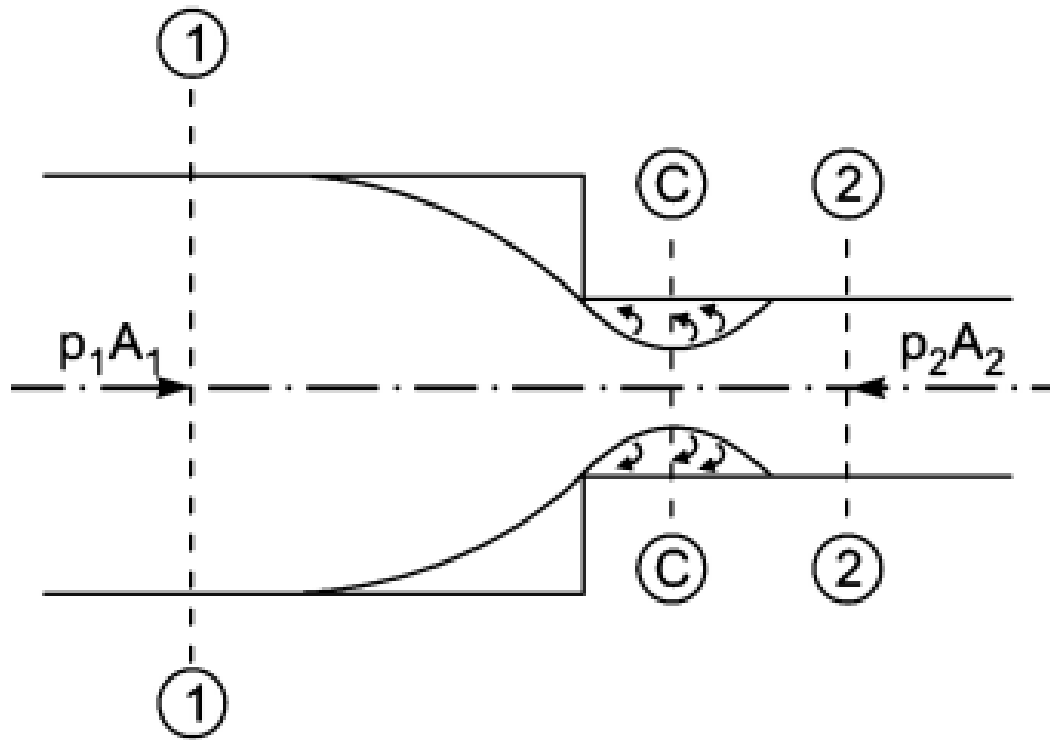
$$h_e = \frac{V_2^2 - V_1 V_2}{g} + \frac{V_1^2}{2g} - \frac{V_2^2}{2g} = \frac{2V_2^2 - 2V_1 V_2 + V_1^2 - V_2^2}{2g}$$

$$= \frac{V_2^2 + V_1^2 - 2V_1 V_2}{2g} = \left(\frac{V_1 - V_2}{2g} \right)^2$$

$$h_e = \frac{(V_1 - V_2)^2}{2g}$$

Minor losses

➤ Losses due to sudden contraction



$$h_c = \frac{V_2^2}{2g} \left[\frac{1}{C_c} - 1 \right]^2$$

$$h_c = 0.5 \frac{V_2^2}{2g}$$

Minor losses

- Head loss at entrance of pipe

$$h_i = 0.5 \frac{V^2}{2g} \quad (\equiv \text{Sudden contraction})$$

- Head loss at exit of pipe

$$h_o = \frac{V^2}{2g} \quad (\equiv \text{Sudden enlargement})$$

Minor losses

- Loss due to pipe bend and pipe fittings

$$h = \frac{kV^2}{2g}$$

Valves and Fittings	Opening	K coefficient
90° elbow standard	-	0.75
90° elbow long radius	-	0.45
45° elbow standard	-	0.35
45° elbow long radius	-	0.2
Coupling / Union	-	0.04
Gate Valve	Open	0.17
	³ / ₄ Open	0.9
	¹ / ₂ Open	4.5
	¹ / ₄ Open	24
Diaphragm Valve	Open	2.3
	³ / ₄ Open	2.6
	¹ / ₂ Open	4.3
	¹ / ₄ Open	21
Globe valve – plug disk	Open	9
	³ / ₄ Open	13
	¹ / ₂ Open	36
	¹ / ₄ Open	112
Angle Valve	Open	2 to 4
Ball Valve	Open	0
	Close 5°	0.05
	Close 10°	0.29
	Close 20°	1.56
	Close 40°	17.3
	Close 60°	206
Butterfly Valve	Open	0
	Close 5°	0.24
	Close 10°	0.52
	Close 20°	1.54
	Close 40°	10.8
	Close 60°	118
Swing check valve	-	2



THANK YOU